Planar Dynamics of a Cube Actuated by a Gimbaled Flywheel

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Technical report

Abstract

This report presents a reduced mathematical model for the planar tipping motion of a cube actuated by a single gimbaled flywheel. Component mass distributions are used to construct effective inertias, gravitational potential energy and gyroscopic coupling terms. Lagrange's equations yield a two-coordinate nonlinear model at prescribed flywheel spin speed. The report also describes the spin-up model, saturated gimbal tracking law, contact treatment and numerical formulation documented in the source implementation. A baseline calculation illustrates the relationship between angular momentum, gravitational resistance and gimbal rate. The formulation reproduces the stated computational model; its geometric approximations, contact assumptions and actuator representations require further verification before experimental predictive accuracy can be established.

1 System definition and modelling assumptions

The system comprises a rigid cube containing a flywheel mounted in a motor-driven gimbal. During a face-to-face tipping manoeuvre, the cube rotates about a ground-contact edge. In the planar representation, this edge is treated as a fixed pivot P. Gimbal motion redirects the flywheel angular momentum and generates a reaction torque about the tipping axis.

The generalised coordinates and state vector are

$$q = \begin{bmatrix} \theta & \gamma \end{bmatrix}^{\mathsf{T}}, \qquad x = \begin{bmatrix} \theta & \dot{\theta} & \gamma & \dot{\gamma} \end{bmatrix}^{\mathsf{T}}. \tag{1}$$

Here, θ denotes cube rotation and γ denotes gimbal rotation relative to the cube. The flywheel spin angle ψ is excluded from the four-state representation; its speed $\Omega = \dot{\psi}$ is prescribed separately. Angular quantities are expressed in radians unless otherwise specified.

The model assumes rigid components, a fixed pivot during free tipping, a diagonal two-coordinate mass matrix, constant gimbal-coordinate inertia and negligible friction in the cube and gimbal coordinates. Tipping integration begins after flywheel spin-up, with constant spin speed during the manoeuvre. Ground contact is represented by separate supported-motion and terminal-impact conditions.

Although cube motion is planar, the gyroscopic actuation mechanism is spatial. The principal tipping input in this formulation arises from reorientation of a spinning flywheel, corresponding to control-moment gyroscope action. It is distinct from the torque generated by changing the speed of a fixed-axis reaction wheel.

2 Coordinates and mass distribution

2.1 Coordinate convention

Let world X point horizontally to the right and Z vertically upwards. Positive θ corresponds to clockwise rotation in the side view. The perpendicular y axis completes a right-handed frame. A cube-fixed point (x, z) maps to world coordinates according to

$$\begin{bmatrix} X \\ Z \end{bmatrix} = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x \\ z \end{bmatrix}. \tag{2}$$

At $\theta = 0$, the pivot is the lower-right corner and the cube's geometric centre is $r_C = (-a/2, a/2)^\mathsf{T}$, where a is the cube side length. The adjacent face reaches the floor at $\theta = \pi/2$.

2.2 Centre of mass

Denote the cage, wheel, brushless motor, gimbal and servo masses by m_c, m_w, m_b, m_g, m_s, respectively. Their total is

$$M = m_c + m_w + m_b + m_g + m_s. \tag{3}$$

The lumped model places each component centre at C, except the servo, which has offset $s = (s_x, s_z)^\mathsf{T} = (0, a/2)^\mathsf{T}$. Consequently,

$$r_G = \frac{(M - m_s)r_C + m_s(r_C + s)}{M} = r_C + \frac{m_s}{M}s, \tag{4}$$

$$x_p = -\frac{a}{2}, \qquad z_p = \frac{a}{2} + \frac{m_s}{M}\frac{a}{2}, \tag{5}$$

$$X_G = x_p\cos\theta + z_p\sin\theta, \qquad Z_G = -x_p\sin\theta + z_p\cos\theta. \tag{6}$$

The brushless motor offset used in the gimbal inertia is not included in this centre-of-mass calculation. The resulting representation is a lumped approximation rather than a fully consistent CAD-based assembly model. In a more general geometry, an offset rotating assembly can introduce gimbal-dependent centre-of-mass terms and additional coupling.

3 Component inertias

3.1 Rotational inertia and axis translation

For a rigid body rotating about a fixed axis,

$$I = \int r_\perp^2 \,\mathrm{d}m, \qquad T = \frac{1}{2}I\omega^2. \tag{7}$$

The spatial representation is

$$\mathbf{I} = \int [(r \cdot r)\mathbf{1} - rr^\mathsf{T}] \,\mathrm{d}m, \qquad T = \frac{1}{2}\omega^\mathsf{T}\mathbf{I}\omega. \tag{8}$$

For parallel axes separated by perpendicular distance d, the parallel-axis theorem gives $I_P = I_C + md^2$. It follows by expanding the squared distance in the inertia integral; the first moments about the centre of mass vanish.

3.2 Annular flywheel

For uniform density ρ, outer radius R_o, inner radius R_i and thickness t_w, integration over concentric rings gives

$$m_w = \int_{R_i}^{R_o} 2\pi\rho t_w r \, \mathrm{d}r = \pi\rho t_w (R_o^2 - R_i^2), \tag{9}$$

$$J_{ws} = \int_{R_i}^{R_o} 2\pi\rho t_w r^3 \, \mathrm{d}r = \frac{\pi\rho t_w}{2}(R_o^4 - R_i^4) = \frac{1}{2} m_w (R_o^2 + R_i^2), \tag{10}$$

$$t_w = \frac{2J_{ws}}{\pi\rho(R_o^4 - R_i^4)}, \tag{11}$$

$$J_{wt} = \frac{1}{2} J_{ws} + \frac{m_w t_w^2}{12} = \frac{m_w}{12}\left[3(R_o^2 + R_i^2) + t_w^2\right]. \tag{12}$$

Subscripts s and t denote spin-axis and transverse inertias. The finite-thickness contribution in J_{wt} is retained; the planar perpendicular-axis relation alone is insufficient for a thick rotor.

3.3 Gimbal and motor representations

The gimbal is approximated as a thin annular ring, giving

$$J_{gs} = \frac{1}{2} m_g (R_{go}^2 + R_{gi}^2), \qquad J_{gt} = \frac{1}{4} m_g (R_{go}^2 + R_{gi}^2). \tag{13}$$

The brushless motor housing is approximated as a uniform solid cylinder of radius r_b and length ℓ_b:

$$J_{bs} = \frac{1}{2} m_b r_b^2, \qquad J_{bt} = \frac{m_b}{12}(3r_b^2 + \ell_b^2). \tag{14}$$

The implemented servo housing contribution is $J_{s,\mathrm{body}} = m_s r_s^2/2$. This expression assumes alignment of the relevant axis with the cylinder symmetry axis; a transverse orientation requires a different inertia expression.

Housing inertia and rotor inertia represent different physical contributions. The brushless rotor inertia J_{br} enters the spin-up calculation. For a transmission ratio $N = \omega_m/\omega_{\mathrm{out}}$, equality of kinetic energies gives

$$\frac{1}{2} J_m \omega_m^2 = \frac{1}{2}(N^2 J_m)\omega_{\mathrm{out}}^2, \qquad J_{\mathrm{reflected}} = N^2 J_m. \tag{15}$$

The baseline reflected servo inertia assumes that the quoted rotor inertia is motor-side. An inertia already referred to the output shaft must not be multiplied by N^2 again.

3.4 Cage inertia

The implemented central cage inertia is $I_{c,C} = 5m_c a^2/18$. This coefficient corresponds to an ideal hollow cube comprising six thin uniform square plates. With plate mass $m_f = m_c/6$, the two plates perpendicular to the central rotation axis each contribute $m_f a^2/6$, while the remaining four each contribute $m_f a^2/3$. Thus,

$$I_{c,C} = 2\frac{m_f a^2}{6} + 4\frac{m_f a^2}{3} = \frac{5}{18} m_c a^2. \tag{16}$$

This derivation explains the implemented coefficient without establishing that the physical frame has the same distribution. A solid cube gives $m_c a^2/6$, while twelve equal thin edge rods give $7m_c a^2/18$. CAD-derived or measured inertia is required for the manufactured structure.

4 Effective coordinate inertias

The geometric centre lies at squared distance $a^2/2$ from the pivot. Defining $d_s^2 = (s_x - a/2)^2 + (s_z + a/2)^2$, the fixed contribution is

$$I_{\text{fix}} = I_{c,C} + (m_c + m_w + m_b + m_g)\frac{a^2}{2} + J_{s,\text{body}} + m_s d_s^2. \tag{17}$$

The central wheel, gimbal and brushless housing inertias are included through their orientation-dependent projections. For an axisymmetric body with unit symmetry axis s,

$$\mathbf{I} = J_t \mathbf{1} + (J_s - J_t) s s^{\mathsf{T}}, \qquad e_y^{\mathsf{T}} \mathbf{I} e_y = J_t + (J_s - J_t)(e_y \cdot s)^2. \tag{18}$$

Choosing $s \cdot e_y = -\sin\gamma$ and defining

$$J_t = J_{wt} + J_{gt} + J_{bt}, \qquad J_s = J_{ws} + J_{gs} + J_{bs}, \tag{19}$$

the effective cube-coordinate inertia and its angular derivative are

$$A(\gamma) = I_{\text{fix}} + J_t \cos^2\gamma + J_s \sin^2\gamma, \tag{20}$$

$$A'(\gamma) = (J_s - J_t)\sin(2\gamma), \qquad \dot{A} = A'\dot{\gamma}. \tag{21}$$

The constant gimbal-coordinate inertia used in the implementation is

$$B = J_{gs} + J_{wt} + J_{sr} + J_{bt} + m_b d_b^2, \tag{22}$$

where J_{sr} is reflected servo rotor inertia and d_b is the motor offset.

These expressions reproduce the reported implementation. The gimbal principal-axis assignments and its axial contribution to B require verification against the assembly geometry. Summing individually transformed inertia tensors and translational contributions may produce a nonzero cross-inertia $M_{\theta\gamma}$ absent from the present formulation.

5 Gravitational potential and tipping threshold

The potential energy and corresponding generalised gravitational torque are

$$U(\theta) = Mg(-x_p \sin\theta + z_p \cos\theta), \tag{23}$$

$$\tau_g(\theta) = -U'(\theta) = Mg(x_p \cos\theta + z_p \sin\theta). \tag{24}$$

Let $r = \sqrt{x_p^2 + z_p^2}$ and $\theta_b = \operatorname{atan2}(-x_p, z_p)$. Then

$$U = Mgr\cos(\theta - \theta_b), \qquad \tau_g = Mgr\sin(\theta - \theta_b). \tag{25}$$

At θ_b, the centre of mass lies above the pivot and potential energy is maximal. This is an unstable balance configuration. The energy barrier from the initial face is

$$\Delta U = Mg(r - z_p). \tag{26}$$

Before this configuration, gravity opposes positive tipping; beyond it, gravity assists rotation towards the adjacent face. Exceeding the energy barrier is necessary but does not alone establish actuator or contact feasibility.

6 Gyroscopic coupling and equations of motion

6.1 Angular momentum and reduced Lagrangian

At wheel speed n in revolutions per minute,

$$\Omega = \frac{2\pi n}{60}, \qquad H = J_{ws}\Omega, \qquad E_{\text{spin}} = \frac{1}{2}J_{ws}\Omega^2. \tag{27}$$

The body equations use wheel-only momentum H, whereas spin-up uses $J_{ws} + J_{br}$. Including motor rotor momentum consistently requires confirmation of spin-axis alignment.

With $H_y = -H\sin\gamma$ and constant H, the flywheel momentum derivative is $\dot{H}_y = -H\dot{\gamma}\cos\gamma$. The corresponding reaction torque on the cube is $H\dot{\gamma}\cos\gamma$. Its projection vanishes at $\gamma = \pm\pi/2$.

For an ideal axisymmetric flywheel, the absolute axial angular velocity is $\dot{\psi} - \dot{\theta}\sin\gamma$. Its kinetic energy is

$$T_w = \tfrac{1}{2}J_{wt}(\dot{\gamma}^2 + \dot{\theta}^2\cos^2\gamma) + \tfrac{1}{2}J_{ws}(\dot{\psi} - \dot{\theta}\sin\gamma)^2 \tag{28}$$

$$= \tfrac{1}{2}(J_{wt}\cos^2\gamma + J_{ws}\sin^2\gamma)\dot{\theta}^2 + \tfrac{1}{2}J_{wt}\dot{\gamma}^2 + \tfrac{1}{2}J_{ws}\dot{\psi}^2 - J_{ws}\dot{\psi}\sin\gamma\dot{\theta}. \tag{29}$$

At prescribed relative spin $\dot{\psi} = \Omega$, the constant spin-energy term can be omitted from the coordinate equations. The reduced Lagrangian is

$$L = \tfrac{1}{2}A(\gamma)\dot{\theta}^2 + \tfrac{1}{2}B\dot{\gamma}^2 - H\sin\gamma\dot{\theta} - U(\theta). \tag{30}$$

This expression reproduces the reduced dynamics; it does not establish that all lumped geometric approximations constitute a single exact rigid-body assembly.

6.2 Lagrange equations

For each generalised coordinate,

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}_i}\right) - \frac{\partial L}{\partial q_i} = Q_i, \qquad Q_\theta = 0, \quad Q_\gamma = \tau_s. \tag{31}$$

For the cube coordinate,

$$\frac{\partial L}{\partial \dot{\theta}} = A\dot{\theta} - H\sin\gamma, \tag{32}$$

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{\theta}} = A\ddot{\theta} + A'\dot{\gamma}\dot{\theta} - H\dot{\gamma}\cos\gamma, \qquad \frac{\partial L}{\partial \theta} = -U', \tag{33}$$

$$\ddot{\theta} = \frac{\tau_g + H\dot{\gamma}\cos\gamma - A'\dot{\gamma}\dot{\theta}}{A}. \tag{34}$$

For the gimbal coordinate,

$$\frac{\partial L}{\partial \dot{\gamma}} = B\dot{\gamma}, \qquad \frac{\partial L}{\partial \gamma} = \tfrac{1}{2}A'\dot{\theta}^2 - H\dot{\theta}\cos\gamma, \tag{35}$$

$$\ddot{\gamma} = \frac{\tau_s - H\dot{\theta}\cos\gamma + \tfrac{1}{2}A'\dot{\theta}^2}{B}. \tag{36}$$

The factor 1/2 follows directly from differentiation of the kinetic energy. In matrix form,

$$\begin{bmatrix} A & 0 \\ 0 & B \end{bmatrix}\begin{bmatrix} \ddot{\theta} \\ \ddot{\gamma} \end{bmatrix} + \begin{bmatrix} A'\dot{\gamma}\dot{\theta} \\ -\frac{1}{2}A'\dot{\theta}^2 \end{bmatrix} + \begin{bmatrix} 0 & -H\cos\gamma \\ H\cos\gamma & 0 \end{bmatrix}\begin{bmatrix} \dot{\theta} \\ \dot{\gamma} \end{bmatrix} + \begin{bmatrix} U' \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ \tau_s \end{bmatrix}. \tag{37}$$

The skew-symmetric gyroscopic matrix transfers power between coordinates without contributing net power to the reduced system.

6.3 Variable spin speed

If H varies with time, the cube equation additionally contains $\dot{H}\sin\gamma$:

$$A\ddot{\theta} = \tau_g + H\dot{\gamma}\cos\gamma + \dot{H}\sin\gamma - A'\dot{\gamma}\dot{\theta}. \tag{38}$$

The reported derivative function omits this term, although it accepts changing rpm. The runner begins tipping at constant spin speed, for which $\dot{H} = 0$. A dynamic spin coordinate would instead require

$$p_\psi = J_{ws}(\dot{\psi} - \dot{\theta}\sin\gamma), \qquad \dot{p}_\psi = \tau_{\text{spin}} - \tau_{\text{loss}}. \tag{39}$$

Prescribed relative speed and conserved absolute spin-axis momentum are different modelling assumptions and must not be interchanged.

7 Energy consistency and local dynamics

The reduced energy is

$$E = \dot{\theta}\frac{\partial L}{\partial \dot{\theta}} + \dot{\gamma}\frac{\partial L}{\partial \dot{\gamma}} - L = \tfrac{1}{2}A\dot{\theta}^2 + \tfrac{1}{2}B\dot{\gamma}^2 + U. \tag{40}$$

Differentiating and substituting the coordinate equations gives

$$\dot{E} = A\dot{\theta}\ddot{\theta} + \tfrac{1}{2}A'\dot{\gamma}\dot{\theta}^2 + B\dot{\gamma}\ddot{\gamma} + U'\dot{\theta} = \tau_s\dot{\gamma}. \tag{41}$$

At zero actuator torque and constant spin speed, reduced energy is conserved during free motion. This is an algebraic and numerical consistency condition, not a statement about total physical energy: the latter also includes flywheel spin, cross terms and spin motor work. Further limiting checks are disappearance of gyroscopic coupling at $H = 0$ and disappearance of A' at $J_s = J_t$. A stationary gimbal can still require holding torque.

Linearisation near $(\theta_b, 0, \gamma_0, 0)$, with $A_0 = A(\gamma_0)$ and $k = H\cos\gamma_0$, yields

$$A_0\delta\ddot{\theta} = Mgr\delta\theta + k\delta\dot{\gamma}, \qquad B\delta\ddot{\gamma} = \delta\tau_s - k\delta\dot{\theta}. \tag{42}$$

An ideally locked gimbal gives $\delta\ddot{\theta} = (Mgr/A_0)\delta\theta$, demonstrating local instability. A free gimbal changes the coupled stability problem. The linear approximation does not describe a complete tipping manoeuvre or impact.

8 Actuator models and numerical implementation

8.1 Flywheel spin-up

The spin-up model uses $J_{\text{spin}} = J_{ws} + J_{br}$, constant available torque τ_m and optional viscous friction coefficient b:

$$J_{\text{spin}}\dot{\Omega} = \tau_m - b\Omega. \tag{43}$$

For zero friction and a stationary initial condition,

$$\Omega(t) = \frac{\tau_m}{J_{\text{spin}}}(t - t_0), \qquad t_{\text{up}} = \frac{J_{\text{spin}}\Omega_{\text{target}}}{\tau_m}. \tag{44}$$

For $b > 0$,

$$\Omega(t) = \frac{\tau_m}{b}\left[1 - e^{-b(t-t_0)/J_{\text{spin}}}\right], \qquad t_{\text{up}} = -\frac{J_{\text{spin}}}{b}\ln\left(1 - \frac{b\Omega_{\text{target}}}{\tau_m}\right). \tag{45}$$

A finite rise time requires $\Omega_{\text{target}} < \tau_m/b$. The reported runner uses $b = 0$ and caps speed at the target, implicitly assuming speed regulation thereafter.

8.2 Gimbal tracking

For desired angle γ_d, let $e = \gamma_d - \gamma$ and $a_d = \omega_n^2 e - 2\omega_n \dot{\gamma}$. The unsaturated command is

$$\tau_{\text{req}} = H\dot{\theta}\cos\gamma - \tfrac{1}{2}A'\dot{\theta}^2 + Ba_d. \tag{46}$$

Under exact model cancellation and a constant target, $\ddot{\gamma} = a_d$ and

$$\ddot{e} + 2\omega_n \dot{e} + \omega_n^2 e = 0. \tag{47}$$

The implemented frequency and limits are

$$\Delta\gamma = \max(|\gamma_d - \gamma_0|, \pi/180), \tag{48}$$

$$\omega_n = 2.5 \min\left(\sqrt{\frac{\tau_{\max}}{B\Delta\gamma}}, \frac{\dot{\gamma}_{\max}}{\Delta\gamma}\right), \tag{49}$$

$$\tau_s = \text{clip}(\tau_{\text{req}}, -\tau_{\max}, \tau_{\max}), \qquad \dot{\gamma}_{\max} = \frac{2\pi n_{s,\max}}{60}. \tag{50}$$

This is a critically damped tracking law only in the ideal unsaturated model. Saturation prevents exact cancellation. The frequency expression is a tuning heuristic; it does not guarantee satisfaction of the speed limit. Overspeed terminates the simulation rather than activating a physical speed limiter. Mechanical servo power is $P_s = \tau_s \dot{\gamma}$ and may be negative during braking.

8.3 State-space integration

With $x = (x_1, x_2, x_3, x_4)^{\mathsf{T}} = (\theta, \dot{\theta}, \gamma, \dot{\gamma})^{\mathsf{T}}$,

$$\dot{x} = \begin{bmatrix} x_2 \\ [\tau_g(x_1) + Hx_4\cos x_3 - A'(x_3)x_4x_2]/A(x_3) \\ x_4 \\ [\tau_s - Hx_2\cos x_3 + \frac{1}{2}A'(x_3)x_2^2]/B \end{bmatrix}. \tag{51}$$

The source reports adaptive RK45 integration with relative tolerance 10^{-7}, absolute tolerance 10^{-9} and maximum step 0.005 s. Events identify contact and gimbal overspeed. The source also reports a fixed-spin energy-identity check at 100 sampled states, with maximum absolute power residual 2.3×10^{-16} W. This reported check supports internal consistency; it is not experimental validation and was not independently rerun for this editorial revision.

9 Ground contact and manoeuvre completion

The smooth equations govern free tipping between supported faces. In supported mode, floor contact enforces $\dot{\theta} = \ddot{\theta} = 0$ while gimbal motion remains possible. At the lower face, departure occurs when unconstrained acceleration becomes positive. At rest,

$$\tau_g(0) + H\dot{\gamma}\cos\gamma > 0. \tag{52}$$

For $H > 0$ and $\cos\gamma > 0$, the corresponding condition is

$$\dot{\gamma} > \frac{-\tau_g(0)}{H\cos\gamma}. \tag{53}$$

At the upper face, the allowable departure direction is negative. A small positive acceleration threshold is used for numerical switching. Departure from a supported face denotes initiation

of edge rotation, not airborne translation. The model does not verify compressive edge reaction or sufficient friction to prevent slip.

The contact loop terminates at the first impact following free motion. An upper-face impact records landing time and incoming angular velocity; rebound, settling and subsequent manoeuvres are not integrated. Although a support-state helper resets cube angular velocity, the terminal impact returns the incoming event state. An optional perfectly inelastic reset $\dot{\theta}^+ = 0$, with γ and $\dot{\gamma}$ unchanged, would remove reduced energy $A(\gamma)(\dot{\theta}^-)^2/2$. This is an idealisation rather than a measured impact law. Performance assessment should therefore distinguish crossing the balance angle from reaching the adjacent face and should report incoming landing speed as well as time.

10 Baseline numerical example

The following values are retained from the source document's baseline parameter set. The selected 6000 rpm operating point is illustrative and is not an optimisation result.

Quantity	Symbol	Value
Cube size; cage mass	$a; m_c$	$0.15\,\mathrm{m};\ 0.50\,\mathrm{kg}$
Wheel radii	$R_o; R_i$	$0.063;\ 0.045\,\mathrm{m}$
Wheel thickness; density	$t_w; \rho$	$0.0115\,\mathrm{m};\ 2700\,\mathrm{kg/m^3}$
Gimbal mass; radii	$m_g; R_{go}; R_{gi}$	$0.20\,\mathrm{kg};\ 0.070;\ 0.065\,\mathrm{m}$
Motor mass; radius; length	$m_b; r_b; \ell_b$	$0.071\,\mathrm{kg};\ 0.016;\ 0.0187\,\mathrm{m}$
Brushless rotor inertia	J_{br}	$3.07 \times 10^{-6}\,\mathrm{kg\,m^2}$
Servo mass; radius	$m_s; r_s$	$0.190\,\mathrm{kg};\ 0.0265\,\mathrm{m}$
Reflected servo inertia	J_{sr}	$9.735 \times 10^{-4}\,\mathrm{kg\,m^2}$
Servo torque; speed bounds	$\tau_{\max}; n_{s,\max}$	$1.3\,\mathrm{N\,m};\ 370\,\mathrm{rpm}$

Substitution gives

$$m_w = 0.189630\,\mathrm{kg}, \qquad J_{ws} = 5.68322 \times 10^{-4}\,\mathrm{kg\,m^2}, \tag{54}$$

$$J_{wt} = 2.86251 \times 10^{-4}\,\mathrm{kg\,m^2}, \qquad M = 1.150630\,\mathrm{kg}, \tag{55}$$

$$(x_p, z_p) = (-0.075, 0.0873845)\,\mathrm{m}, \qquad I_{\mathrm{fix}} = 0.0193426\,\mathrm{kg\,m^2}, \tag{56}$$

$$B = 0.00218596\,\mathrm{kg\,m^2}, \qquad A(0) = 0.0200917\,\mathrm{kg\,m^2}, \tag{57}$$

$$A(\pi/2) = 0.0208325\,\mathrm{kg\,m^2}. \tag{58}$$

Thus $A(\gamma) = 0.0200917 + 0.000740796\sin^2\gamma$ in $\mathrm{kg\,m^2}$. The balance and momentum quantities are

$$\theta_b = 40.6387^\circ, \qquad \Delta U = 0.313483\,\mathrm{J}, \tag{59}$$

$$\Omega = 628.319\,\mathrm{rad/s}, \qquad H = 0.357087\,\mathrm{kg\,m^2/s}, \tag{60}$$

$$E_{\mathrm{spin}} = 112.182\,\mathrm{J}. \tag{61}$$

At $\theta = \gamma = \dot{\theta} = 0$, gravity contributes $-0.846576\,\mathrm{N\,m}$, giving the local critical gimbal rate

$$\dot{\gamma}_{\mathrm{crit}} = \frac{0.846576}{0.357087} = 2.37078\,\mathrm{rad/s} = 22.64\,\mathrm{rpm}. \tag{62}$$

For $\dot{\gamma} = 3\,\mathrm{rad/s}$ at this configuration,

$$\ddot{\theta} = \frac{-0.846576 + 0.357087(3)}{0.0200917} \approx 11.18\,\mathrm{rad/s^2}. \tag{63}$$

This is an instantaneous acceleration rather than proof of manoeuvre completion. At $\gamma = 60°$, the projected gyroscopic leverage is halved, requiring twice the gimbal rate to oppose the same gravitational torque. With spin torque $0.0985\,\mathrm{N\,m}$ and zero viscous friction,

$$t_{\mathrm{up}} = \frac{(5.68322 \times 10^{-4} + 3.07 \times 10^{-6})(628.319)}{0.0985} \approx 3.645\,\mathrm{s}. \tag{64}$$

The stored spin energy exceeds the gravitational barrier, but this comparison alone does not establish successful energy transfer. Torque projection, gimbal rate, actuator limits and contact conditions remain decisive.

11 Limitations and further model development

The reduced equations are internally derivable under the stated assumptions. Their predictive accuracy depends on verification of component inertia tensors, centres of mass, gimbal-axis assignments, gearbox-referred inertia and consistent flywheel-plus-rotor angular momentum. Measured torque–speed characteristics, friction, power limits and contact behaviour are also required.

A more general formulation should construct each body's position $r_i(q)$ and angular velocity $\omega_i(q, \dot{q})$ and assemble

$$T = \sum_i \left[\tfrac{1}{2} m_i \dot{r}_i^{\mathsf{T}} \dot{r}_i + \tfrac{1}{2} \omega_i^{\mathsf{T}} \mathbf{I}_i \omega_i \right], \qquad U = \sum_i m_i g Z_i(q), \tag{65}$$

with consistent frame conventions for the inertia tensors and angular velocities. Lagrange's equations then generate offset, cross-inertia and gimbal-dependent gravitational terms. Floor reaction and impact laws require separate formulation. This provides a systematic basis for refining the planar model and extending the investigation to spatial motion.

12 Implementation provenance

The source document records examination of the local `Gimbaled-Reaction-Cube` repository on 9 November 2026, revision `6f52c0e`, with local modifications present. The implementation statements in this report refer to that recorded snapshot rather than an independently verified current revision. Paths below are relative to `2d/`.

Source file	Model contribution
`hardware/parameters.py`	Baseline geometry, masses and actuator values
`hardware/cube.py`	Centre of mass, effective inertia, gravity and cage inertia
`hardware/flywheel.py`	Annular geometry, momentum and spin energy
`hardware/gimbal.py`	Gimbal inertias and coordinate inertia B
`hardware/brushless_motor.py`	Housing and rotor parameters
`hardware/servo_motor.py`	Servo housing inertia and actuator limits
`dynamics/dynamics_2d.py`	Nonlinear acceleration equations
`dynamics/physical_constraints.py`	Supported motion and terminal impacts
`control/flywheel_control.py`	Analytical spin-up law
`control/gimbal_control.py`	Saturated tracking controller
`simulation/simulation_runner.py`	Integration start and numerical settings